15°. Therefore, when air is filled in the tyre, the bead makes a tight fit with the rim. In tubeless tyres this tilt structure acts as a tight seal.

The disc in the disc wheel does the work of the spoke. The wheel is fitted on the axle by the bolts. There are holes provided on the wheel disc as shown in the picture. Through these holes air is allowed to enter inside the brake drum and the air cools the brake drum. There is a gap provided for the wheel disc and a hole is made on the rim for placing the tube valve.



Figure 5.3.1.1 Rim and Disc

Advantages

1. The construction of the disc wheel is simple, durable and cheap.
2. Its maintenance is low.
3. It is suitable for all light and heavy vehicles.
4. No curves or bends occur on the wheel.

5.3.2 Wired wheel

Unlike the disc wheel, the hub is located at the center of the wheel and the rim is at the top separately. The rim and the hub are connected by the wires called as spokes. The structure of the wired wheel can be seen in Figure 5.3.2. We can find this type of wheels used in two wheelers such as bicycles and motor cycles. The spokes in the two wheels withstand the vehicle weight, pulling force and braking force. For withstanding these forces the spokes are connected between the wheel hub and the rim. There are splines made in the hub for fixing it on the hub axle. The weight of wired wheel is comparatively lower than the disc wheel and has high cooling efficiency and hence it is used in racing cars.



Figure 5.3.2 Structure of a Spoke / Wired Wheel

Advantages of wired wheel

1. It is lower in weight and has more durability than a disc wheel.
2. It allows the brake drum to cool very quickly and easily.
3. It can be fixed and dismantled easily whenever necessary.

Disadvantages

1. Maintenance cost is high.
2. There are possibilities for bends to occur in the wheel.
3. Tubeless tyres cannot be used because of the holes present in the rim for the spoke wires.

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